

The dataset contains 7,043 customers, of which 1,869 customers (26.5%) churned and 5,174 customers (73.5%) remained with the company. The distribution shows that most customers did not churn, indicating an imbalanced dataset. This imbalance should be considered when evaluating model performance because a high accuracy alone may not fully reflect the model's ability to identify churn customers.

Contract type shows the strongest relationship with customer churn. Customers with month-to-month contracts had a churn rate of 42.7%, while one-year and two-year contracts had much lower churn rates of 11.3% and 2.8%, respectively. This suggests that customers with longer contract commitments are more likely to remain with the company, while customers without long-term commitments are more likely to leave.

Customers using Fiber optic internet showed a churn rate of 41.9%, which is considerably higher than DSL customers (19.0%) and customers without internet service (7.4%). This result indicates that churn behavior differs across service types, suggesting that internet service may be an important factor associated with customer retention.

Customers who paid by electronic check had the highest churn rate (45.3%), while customers using automatic payment methods had much lower churn rates. This suggests that payment method is associated with customer churn and may provide useful information for predicting customer behavior.

The boxplots show that customers who churned generally had shorter tenure and higher monthly charges than customers who remained with the company. This

suggests that customers who have recently joined the company or pay relatively high monthly fees may have a higher likelihood of churning.

The Logistic Regression model achieved an overall accuracy of 82.1% on the testing dataset. The confusion matrix indicates that the model correctly classified most non-churn customers while identifying a reasonable number of churn customers.

For the churn class, the model achieved a precision of 0.69, recall of 0.60, and F1-score of 0.64. Although the model performed well overall, the relatively lower recall indicates that some customers who churned were still classified as non-churn.

Overall, the analysis indicates that contract type, tenure, internet service, and payment method are associated with customer churn.